Wilson Wu

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Education

Tufts University Medford, MA
Bachelor of Science in Electrical Engineering, Minor in Computer Science *May 2027*
GPA: 3.87, Dean's List (Fall 2023 – Spring 2026), IEEE-HKN

Technical Experience

Safe & Performant Autonomous Robotics & Control Lab (SPARC), Tufts University Medford, MA
Directed Researcher, advised by Prof. Ryan Cosner *September 2026 – Present*

- Developing field-of-view-aware trajectory planning, based on covariance steering, that keeps safety-critical objects in view for a semantically aware safe controller on a Go2 robot dog

University of Wyoming – Controlled Environment Agriculture NSF-REU Remote
Undergraduate Researcher, advised by Prof. Liping Wang *June – August 2026*

- Built a computer vision pipeline for salt-stress phenotyping in hydroponic tomatoes: YOLO segmentation feeding 42 features, evaluated on 149 plants across 8 salinity levels under grouped cross-validation
- Validated against harvest measurements the model never saw, finding that greenness predicted plant weight only in damaged plants and finding that lighting differences inflated the dose results

Automated Systems and Robotics Lab – Analytical Scene Reconstruction with LiDAR Medford, MA
Undergraduate Researcher, advised by Prof. Jason Rife *January – May 2026*

- Benchmarked six interpolation methods for LiDAR view synthesis on Ouster OS2-128 data from the Volpe National Transportation Center, and designed a new FFT-based method for per-pixel range synthesis
- Found the error is structured rather than random: every method fails in a predictable way at depth edges, placing the methods on an underfitting-to-overfitting spectrum

Tufts University Department of Electrical and Computer Engineering Medford, MA
Teaching Assistant: Embedded Systems *January – May 2026*

- Instructed 80 students in embedded systems design on the STM32L432KC, covering GPIO, timers, interrupts, DMA, and peripheral interfacing in C

Teaching Assistant: Digital Logic *September – December 2025*

- Instructed 80 students in building digital logic on UPduino v3.0 FPGA hardware in SystemVerilog

Projects

Autonomous LiDAR Navigation Robot *January 2026 – Present*
Verified in simulation; hardware bring-up on RPLIDAR A2M12, STM32L432KC, Raspberry Pi 5

- Designing a complete navigation stack from scratch in Python: ES-EKF, ICP, occupancy mapping, A* planning
- Built the scan matcher to report when its surroundings are too featureless to fix a position, so the filter rejects confident-looking but meaningless corrections in bare corridors
- A 200-step simulated run held 6.6 mm position and 3.4 mrad heading error. Work continues into a CBF safety filter that widens when localization is uncertain and tightens it when localization is sharp

Digital Piano

- Developed an RTOS polyphonic digital piano on STM32 with a 48 kHz DMA-driven DAC, double buffering, and analog RC filtering for 5 ms latency audio, plus an ADSR additive synthesizer using fixed-point phase accumulators

Skills

Languages: C, C++, Python, MATLAB, VHDL, LaTeX
Tools: KiCAD, Lattice Radiant, LTSpice, Fusion 360, RoboFlow, YOLO, Linux, Git, VSCode, PlatformIO
Hardware: STM32, ESP32, UPduino FPGA, Raspberry Pi, LiDAR, IMUs